M MOUKTHIKA

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Education

SRM Institute of Science and Technology **Aug 2023 – May 2027**
B.Tech in Computer Science and Engineering ***(CGPA: 9.2/10)*** *Chennai*

Projects

RiskSure | *Python, FastAPI, Pydantic, SQLAlchemy, Scikit-learn, SHAP, React.js, Docker* **[Oct 2026]**

- Built a **full-stack InsurTech platform** modeled on **Guidewire**-style policy systems, with a **config-driven rating engine** where each product (auto, home, small business) is one Pydantic-validated YAML file and needs **zero product-specific code**; premiums are computed deterministically with rounding at every step, so the breakdown (base cover → factors → discounts → loadings) sums exactly to the final premium.
- Trained a **gradient-boosting claim-risk model** (0.94 cross-validated AUC on 5,000 synthetic policies) with **per-applicant SHAP explanations**, and fitted home and business rating factors using a **Tweedie GLM** with credibility blending and caps; routed the ML score through underwriting rules instead of the price to avoid double-counting risk.
- Implemented a **rules engine** (decline > refer > loading > approve) with a **backtest preview** over 9,000 applicants showing impact on decisions, premium and loss ratio, plus an **LLM rule-authoring** flow where Claude emits schema-constrained JSON that is validated, previewed and published only after human approval.
- Enforced the quote lifecycle (Draft → Quoted/Referred/Declined → Bound → Issued) as a **state machine** returning **HTTP 409** on illegal transitions, writing audit events **in the same DB transaction** as each change; covered by **147 pytest tests** and deployed via Docker (Render) with a Vercel frontend.

TrustTrail | *Python, FastAPI, DuckDB, Groq API (Llama 3.3), Sentence Transformers* **[May 2026]**

- Built a conversational analytics platform for financial bank analysts that understands plain-English queries and converts them into SQL, cutting multi-hour data requests to sub-second answers.
- Implemented Gen AI with the Groq API (Llama 3.3) to turn executed NL-to-SQL results into clear, readable English responses.
- Delivered audit-ready answers with full explainability: the data source, the executed SQL and a confidence score shown with every response.
- Reduced LLM hallucinations with a 20-KPI semantic metric dictionary that grounds column names and definitions, a 4-intent query router (keyword rules with MiniLM semantic fallback), and in-memory DuckDB execution so answers come from real query results.
- Ranked **Top 30 among 5,000+ teams** at a national-level fintech hackathon (NatWest Code for Purpose).

OncoVision | *Python, TensorFlow, MobileNetV3, Grad-CAM++, t-SNE, RAG* **[April 2026]**

- Built an explainable MobileNetV3 pipeline classifying histopathology images into three classes with 94% accuracy and F1-score.
- Applied two-phase transfer learning, Bayesian optimization and Focal Loss to handle class imbalance and stabilize training.
- Used t-SNE for feature representation and added Grad-CAM++ heatmaps and a RAG explanation layer, outperforming YOLOv4 and DenseNet201 as the only explainable model.

Technical Skills

Languages: C++, C, Python, JavaScript, SQL
Operating Systems & Networking: Linux, Operating Systems Concepts, Computer Networks (TCP/UDP/IP)
Cloud & As-a-Service: AWS, Cloud Service Models (IaaS, PaaS, SaaS), Docker, CI/CD, Git, GitHub
Database Systems: MySQL, SQLite, DuckDB, SQL Querying
Security & Authentication: Authentication (JWT), Secure REST APIs
Software Development & QA: REST APIs, FastAPI, Flask, React.js, Automated Testing, Debugging
AI/ML: Scikit-learn, TensorFlow, XGBoost, Classification, Regression, Explainable AI (SHAP, Grad-CAM++), Model Evaluation
Data Science: Pandas, NumPy, Data Preprocessing, Exploratory Data Analysis, Feature Engineering, Data Visualization
LLM & Agentic AI: LangChain, LangGraph, RAG, MCP, Tool Calling, Fine-Tuning, Vector Embeddings, LangSmith, LLM-as-a-Judge
Soft Skills: Analytical Thinking, Problem Solving, Team Collaboration

Publications

- **RenalGuard-X: An Explainable Model for Accurate Chronic Kidney Disease Prediction** October 2026
  – Published in *2026 IEEE International Conference on Natural Language Processing and Computer Vision (ICNPCV)*

- **ColoXAI: Explainable Framework for Colorectal Cancer Prediction** August 2026
  – Presented at *2*nd IEEE International Conference on Distributed Systems, Computer Networks and Cybersecurity (ICDSCNC 2026).

Certifications & Achievements

- **AWS Certified Cloud Practitioner** – Amazon Web Services
- **AWS Academy Cloud Foundations** – Amazon Web Services
- **AWS Academy Machine Learning Foundations** – Amazon Web Services
- **Enterprise Cloud Engineering for Insurance Technology** – Guidewire software
- **Microsoft GitHub Copilot** – Microsoft
- **NPTEL** – Machine Learning , Programming in Java
- Participated in **15+** national and university-level hackathons.
- Completed the **AICTE AI-ML Virtual Internship**.
- Solved **100+ problems** on LeetCode.

Leadership / Extracurricular

IEEE SSIT **Aug 2024 – Present**
Corporate Lead *IEEE - SRM*

- Led a team of 35 students and organized the chapter's flagship event, coordinating the team and logistics from planning through execution.

Directorate of Student Affairs **Jan 2024 – Jan 2025**
Public Relations Member *SRM Institute of Science and Technology*

- Managed public relations for **Milan'24**, SRM's national-level fest, and managed multi-university competitions with **7,000+ participants**.